SIRD – Sports Injury Risk Detection from Video

AI-Powered Sports Movement Analysis
Infosys Springboard Internship Project

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Introduction

What is SIRD?

SIRD is an AI-enabled system that analyzes athlete movement from video to detect injury-risk indicators and present role-specific insights.

Why prevention matters

Early identification of risky movement patterns reduces downtime, improves performance continuity and supports evidence-based training and rehab decisions.

Why video-based analysis

Video is ubiquitous, non-invasive, and captures context-rich movement — enabling retrospective review, longitudinal tracking and scalable monitoring.

Problem Statement

Manual assessment limits

Manual video review is time-consuming and inconsistent across evaluators.

Hidden abnormalities

Subtle deviations in kinematics are easy to miss without quantitative measures.

Need for structured monitoring

Athletes require repeatable tracking of movement and recovery progress over time.

Fragmented information

Coaches and support staff lack a centralized, role-based view of analysis and recommendations.

Objectives

Analyze videos

Ingest and validate athlete video uploads for structured processing.

Extract movement features

Derive pose and kinematic features useful for risk assessment.

Detect risk indicators

Flag movement patterns correlated with elevated injury risk.

Maintain profiles

Track athlete history, videos and longitudinal metrics.

Role-based insights

Deliver tailored dashboards for Admin, Athlete, Coach, Physio, Scientist.

Support decisions

Provide actionable guidance for training and recovery planning.

Existing System — Limitations

- Observation-heavy workflows with human variability
- Poor historical tracking and fragmented records
- No centralized repository for multi-session comparisons
- Limited automation for extracting quantifiable movement metrics
- Hard to scale across teams and organizations

SIRD addresses these gaps by introducing automated, consistent video processing and unified role-based reporting.

Proposed System — SIRD Overview

Core user flows and capabilities

- User authentication (JWT)
- Athlete profile & organization management
- Video upload and validation
- AI-driven video processing and risk prediction
- Role-based dashboards and training/recovery guidance

01

1. Upload

Secure video submission by athlete or staff

02

2. Process

Frame-level pose extraction and feature derivation

03

3. Predict

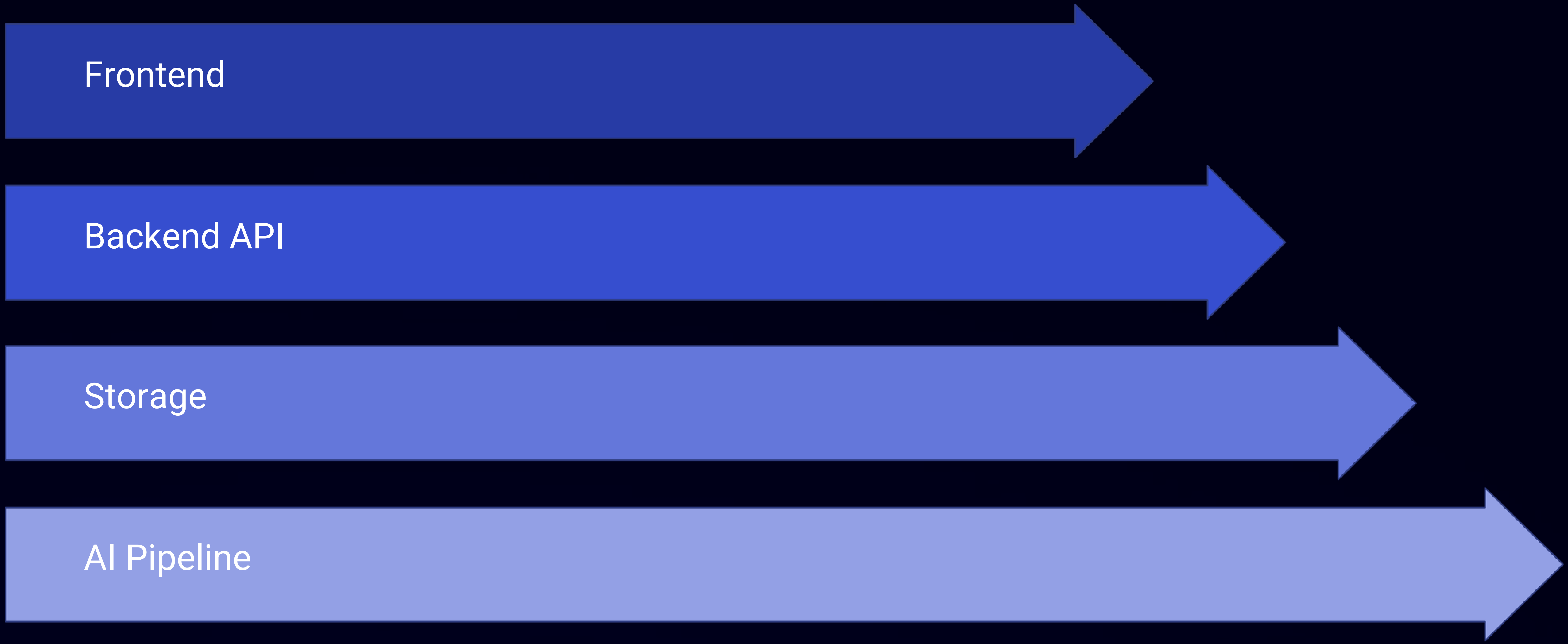
Compute injury-risk indicators and flag concerns

04

4. Report

Deliver role-specific visualizations and guidance

System Architecture



This architecture separates UI, API, persistent storage and the video/AI pipeline to enable scalability and clear responsibility boundaries.

Technology Stack

Frontend

React + Vite — responsive SPA for dashboards and uploads

Backend & AI

Python + FastAPI for APIs and the CV/AI pipeline

Storage

PostgreSQL managed schema via SQLAlchemy

Auth & Dev

JWT authentication; Git/GitHub for source control; VS Code for development

User Roles & Access

Admin

Manage organizations, generate join codes, user and system monitoring

Athlete

Upload videos, view personal progress, follow guidance

Coach

Review athlete performance, quality scores and prescribe training

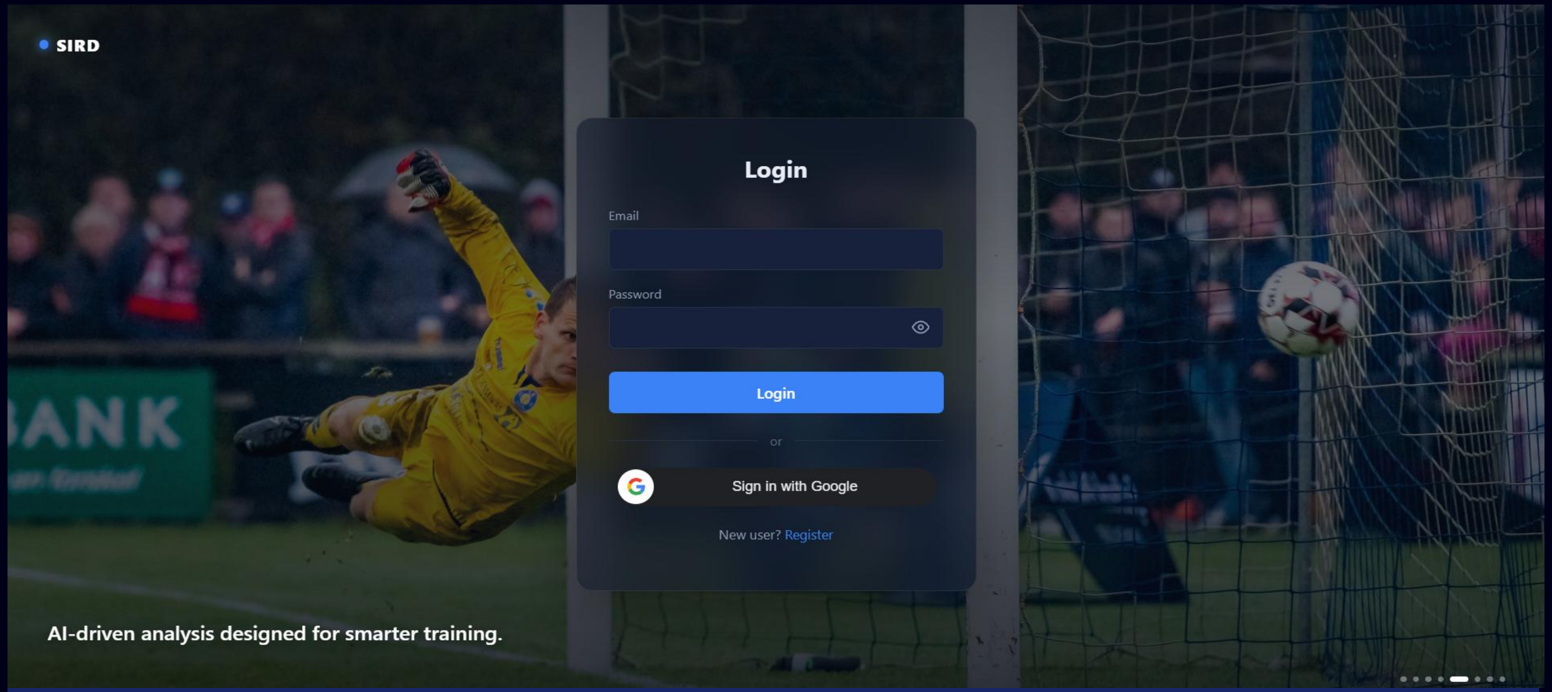
Physiotherapist

Access injury-relevant metrics, recovery plans and session notes

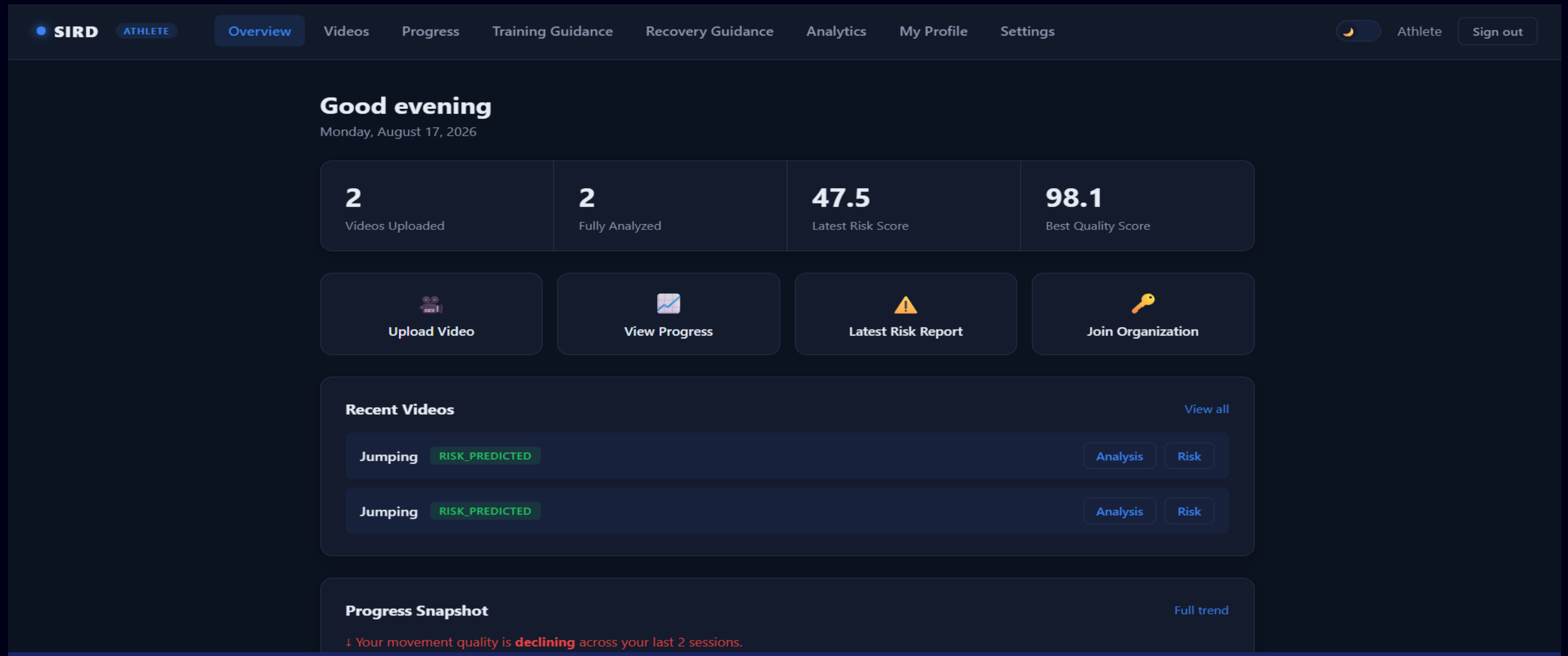
Sports Scientist

Access research-grade analysis, exportable feature data and cohort comparisons

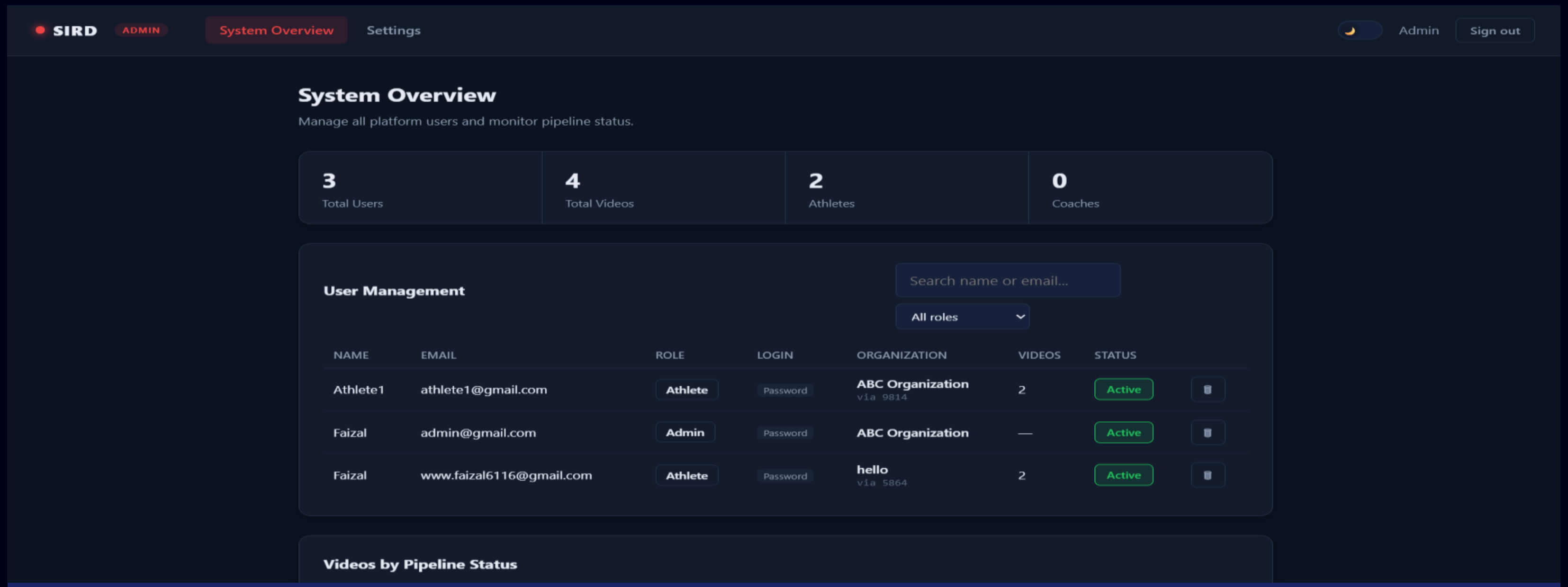
Application Screenshots



Login Page

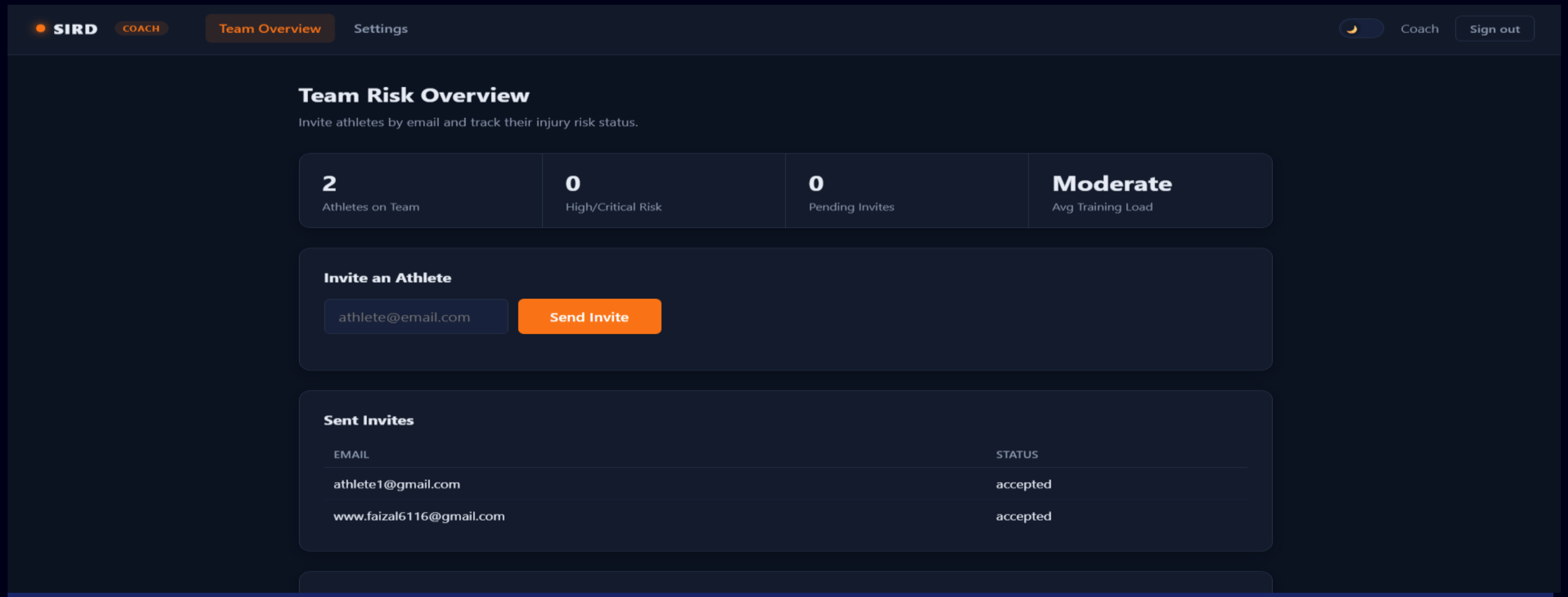


Athlete Dashboard

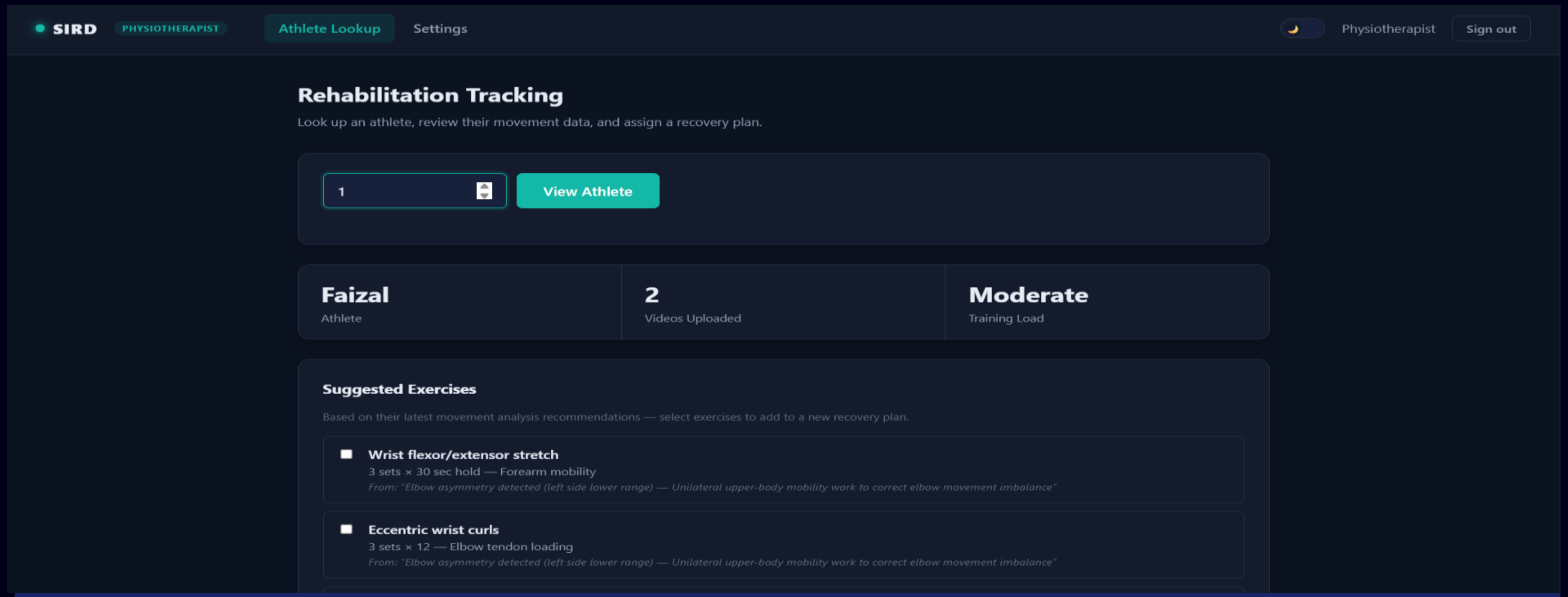


Admin Dashboard

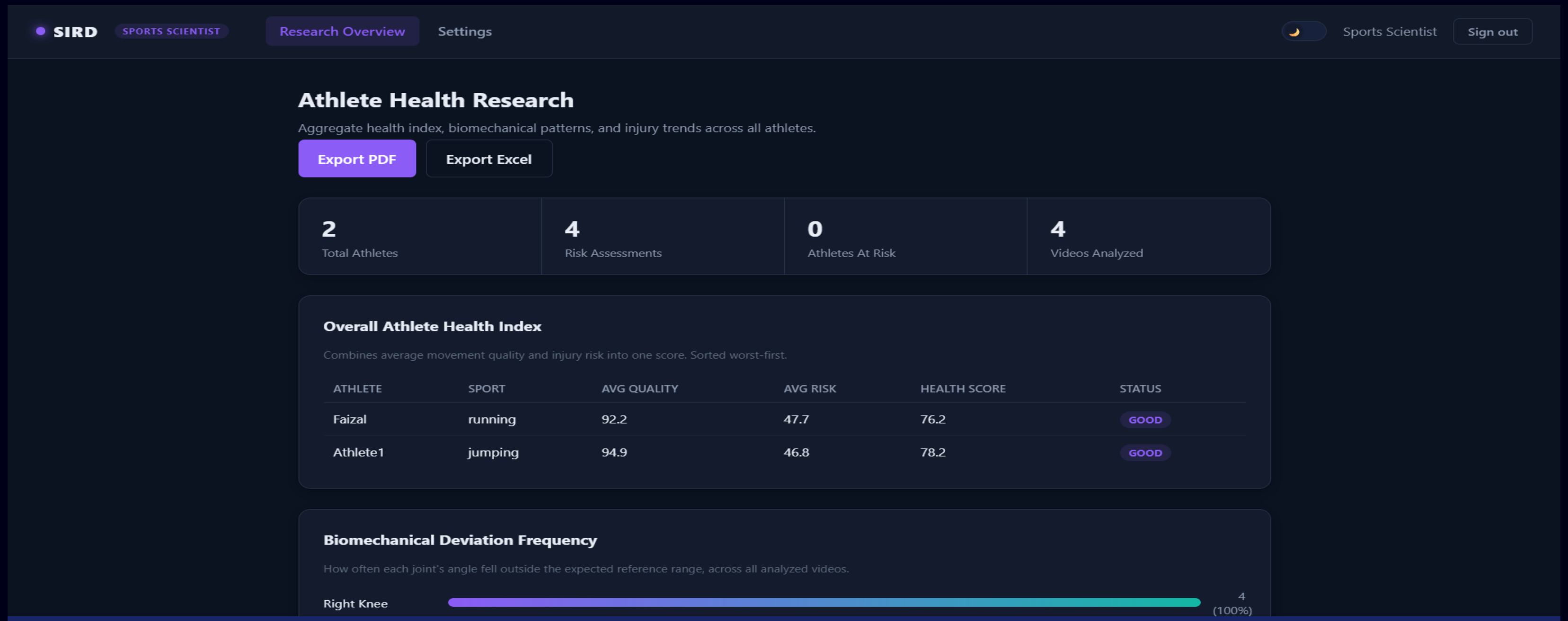
Role-Based Dashboards



Coach Dashboard



Physiotherapist Dashboard



Sports Scientist Dashboard

Future Enhancements

- **Real-Time Video Analysis**
  Enable live movement and injury-risk analysis during training sessions.
- **Advanced AI Models**
  Improve movement analysis and risk prediction using more advanced AI techniques and larger datasets.
- **Mobile Application**
  Provide SIRD features through Android and iOS applications.
- **Cloud Deployment**
  Deploy the system on cloud infrastructure for scalable access and storage.
- **Personalized Training Plans**
  Generate customized training and recovery recommendations based on athlete data.
- **Long-Term Athlete Monitoring**
  Track performance and injury-risk trends over multiple training sessions.

Conclusion

SIRD provides an integrated platform for video-based sports movement analysis and injury-risk assessment.

The system enables athletes to upload and monitor their performance while providing role-based access for coaches, physiotherapists, sports scientists, and administrators.

By combining video processing, AI-based analysis, athlete profiles, risk indicators, and centralized dashboards, SIRD aims to support **early risk identification, informed decision-making, and safer sports training**

Q&A

THANK YOU